

Review of Cross-Chain Implementations

Cross-chain implementations facilitate interoperability among diverse blockchain networks, enabling seamless asset transfers, data sharing, and smart contract execution. Recent scholarly works highlight advancements in protocols, standards, and applications, especially in 2025, with an emphasis on security, scalability, and the adoption of practical applications.

Key protocols encompass **bridges**, **cross-chain messaging**, and **shared validator models**. Bridges, exemplified by those within Chainlink's Cross-Chain Interoperability Protocol (CCIP), facilitate token transfers with embedded instructions for destination-chain actions, leveraging decentralized oracle networks (DONs) to detect anomalies and effectively manage security and risk. CCIP also supports self-service Cross-Chain Tokens (CCTs) for native multichain assets, prioritizing modular security without vendor lock-in. Similarly, the Inter-Blockchain Communication (IBC) protocol developed by Cosmos allows for the exchange of arbitrary messages across different chains, thereby generalizing bridges for trustless operations. Shared validator protocols, as implemented in Polkadot and Cosmos, utilize common validators to enhance efficiency, prevent asset locking, and enable quicker verification processes than those available through traditional method bridges.

Standards such as **IEEE Std 3221.01-2025** establish frameworks for cross-chain transactions, encompassing principles, applications, and performance metrics, thereby fostering interoperability between public and private blockchains. Surveys conducted in 2025, including those focusing on Bitcoin cross-chain bridges, offer taxonomies that classify implementations based on mechanisms such as relays and sidechains, as well as their potential for integration with artificial intelligence. Additionally, a comprehensive survey evaluates various cross-chain solutions, highlighting challenges such as atomicity and security, and notes that projects like Polkadot's relay chain serve as exemplary instances of dynamic virtual interoperability networks.

Applications extend beyond finance: within supply chains, blockchain technology enhances transparency through cross-chain data sharing; in intellectual property trade, it addresses barriers to cross-border enforcement, such as scalability; and in payments, it transforms global transactions by reducing fees. Trends projected for 2025–2030 emphasize hybrid models, the integration of artificial intelligence, and intent-based systems (e.g., Across Protocol) to facilitate user-friendly multi-chain experiences. Challenges include vulnerabilities (e.g., bridge hacks) and the necessity for layered security.

Ethereum-based bridging architectures exemplify practical implementations that interconnect various chains to facilitate seamless operations, though they also underscore scalability trade-offs.

Review of Content Rights Management Literature

Content rights management (CRM), particularly blockchain-based digital rights management (DRM), focuses on safeguarding intellectual property through secure, transparent mechanisms. Recent literature reviews from 2025 highlight decentralization, integration with artificial intelligence, and market growth.

A comprehensive 2025 review of Digital Rights Management (DRM) development traces its evolution from traditional Technological Protection Measures (TPMs) to blockchain-based schemes. This evolution notably emphasizes the use of smart contracts to enable automated

licensing and enforcement. Market analyses forecast that the blockchain DRM sector will expand from USD 0.25 billion in 2025 to USD 1.42 billion by 2029. This growth is primarily driven by the media and entertainment industry's needs for asset protection. Decentralized systems are increasingly supplanting centralized networks through peer-to-peer models, thereby reducing the risk of security breaches and enhancing system availability.

In libraries and digital archives, blockchain technology enhances security by leveraging immutable ledgers for content tracking, complemented by AI integration for efficient rights verification. Design principles for blockchain-based Digital Rights Management (DRM), exemplified by the music industry, advocate for integrated solutions that promote transparent licensing and address inefficiencies inherent in single-chain models. Additionally, cross-industry frameworks such as the Linked Content Coalition (LCC) promote standards-based metadata exchange to facilitate automated rights management across various media platforms sectors.

Legal perspectives examine the role of blockchain in copyright enforcement, utilizing smart contracts for token-based access and decryption; however, challenges such as regulatory alignment remain. Recent developments (July 2025) encompass broader trends in digital assets, including digital rights management (DRM) within blockchain ecosystems. Notable gaps include scalability issues in cross-border intellectual property and the need for hybrid artificial intelligence (AI) and blockchain integration systems.

Cross-chain specific literature, such as a 2024 scheme for digital resource copyright, employs sidechain/relay mechanisms for registration, transactions, modifications, and enforcement, thereby enhancing efficiency compared to single-chain approaches.

How These Pertain to the Project

The project's cross-chain Customer Relationship Management (CRM) service for decentralized subscription, Pay-Per-View (PPV), and purchase models directly builds upon these implementations. Cross-chain protocols such as XCM (native to Polkadot) are aligned with CCIP and IBC to facilitate seamless rights transfers. This alignment enables the integration of the XCM methodology for low-latency transactions across parachains and bridges, including those to Ethereum. The IEEE standards and relevant surveys inform the evaluation of interoperability metrics—such as success rates and latency—addressing challenges such as atomicity in PPV and purchase verification. Shared security models relevant to the project's deployment on Polkadot contribute to enhanced scalability, achieving over 500 transactions per second (TPS) compared to Ethereum-based competitors, as documented in the intellectual property trade barriers.

The existing literature on Customer Relationship Management (CRM) emphasizes prioritizing monetization over mere protection. Reviews of blockchain Digital Rights Management (DRM) confirm the effectiveness of ink! contracts to facilitate automated subscriptions and ownership transfers, thereby addressing gaps in transparent revenue sharing, with fees typically below 5%. Cross-chain copyright schemes directly inform the framework's processes, such as registration via sidechains, thereby overcoming the limitations inherent in single-chain systems and improving real-world applicability. Hybrids of artificial intelligence and blockchain technology offer potential enhancements to the project's evaluation, including the use of fuzzing techniques for security assessments. Market growth projections highlight the importance of economic efficiency.

Implementation Notes (Phase 4–5 Measured Results)

The section above was written during Phase 1 using theoretical projections. Phase 4–5 implementation produced the following measured results that contextualise the literature predictions:

Table 1: Literature Claims vs Measured Results

Literature Claim	Measured Result	Assessment
Polkadot enables “500+ TPS via parachains”	27 TPS sustained per parachain (283 TPS theoretical)	Single-parachain throughput is lower than aggregate network predictions. Polkadot’s scaling model is horizontal (more parachains), not vertical (faster single chain).
“Monte Carlo simulations quantify cost savings of approximately 60%”	Monte Carlo not implemented; centralized benchmark built instead	Comparative analysis shows 270× performance cost for trustlessness. The self-publishing model eliminates intermediary commissions (typically 15–30%) entirely.
ink! contracts for automated subscriptions	Implemented as FRAME pallet + ink! API layer	Architecture pivoted from pure ink! to FRAME pallet for performance. ink! contract provides external interface via pallet-revive precompile.
Cross-chain copyright schemes inform registration	XCM cross-chain operations proven (subscribe, renew, PPV, ownership)	100% success rate, 18–32 second latency via HRMP relay
Snowbridge for Ethereum interoperability	Full Snowbridge v1 E2E demonstrated	2 ETH bridged from local Ethereum → Bridge Hub → AssetHub with cryptographic proof verification

Overall, these developments advance the research question by establishing benchmarks for scalability and interoperability. Meanwhile, the project introduces innovations in Polkadot-specific monetization, thereby bridging gaps in creator empowerment and cross-network collaboration efficiency.

Annotated Bibliography

This annotated bibliography lists 23 references from the cross-chain implementation and content rights management literature, supporting the thesis for a cross-chain CRM service for

decentralized subscription, PPV, and purchase models. The entries are organized thematically around cross-chain interoperability and blockchain-based DRM solutions.

Cross-Chain Implementations

1. **Chainlink. (2021).** Introducing the Cross-Chain Interoperability Protocol (CCIP) for Decentralized Inter-Chain Messaging and Token Movements. <https://blog.chain.link/introducing-the-cross-chain-interoperability-protocol-ccip/>

Annotation: The paper explains the rationale behind Chainlink’s optimization as a generalized cross-chain communication protocol within the blockchain industry. It further details the integration of CCIP within a multilayered technological framework and discusses how CCIP will enable diverse services to advance cross-chain hybrid smart contracts.

2. **Sevim, H.S. (2024).** A Survey on Trustless Cross-chain Interoperability Solutions in On-chain Finance. *DLT 2024 Proceedings*. https://dlt2024.di.unito.it/wp-content/uploads/2024/05/DLT2024_paper_14.pdf

Annotation: This document analyzes fundamental principles, current advancements, and innovative examples of cross-chain interoperability protocols. It compares leading trustless protocols—LayerZero, Wormhole, Chainlink, Circle, Polkadot, and Cosmos—focusing on their design, mechanisms, consensus, and limitations.

3. Mao, H., et al. (2022). A Survey on Cross-Chain Technology: Challenges, Development, and Prospect. *ResearchGate*. <https://www.researchgate.net/publication/366226788>

Annotation: This paper reviews cross-chain methods from a technical, security, and limitations perspective, assessing their applicability. It discusses challenges faced by public and consortium blockchains and introduces the main approaches used. The review covers eight key projects and their applications, ending with future research directions.

4. **Tang, G., et al. (2025).** Bitcoin Cross-Chain Bridge: A Taxonomy and Its Promise in Artificial Intelligence of Things. *arXiv*. <https://arxiv.org/abs/2509.10413>

Annotation: Classifies cross-chain bridges, relevant to the project’s bridge design for external chains such as Ethereum. Suggests AI integration for future extensions but focuses on Bitcoin, limiting direct applicability to Polkadot-based content rights.

5. **Valaštín, V., et al. (2024).** Protocol for unifying cross-chain liquidity on Polkadot. <https://pdfs.semanticscholar.org/5152/064e23fb989674640caa438495dbb83a88c.pdf>

Annotation: Describes a protocol for cross-chain liquidity, applicable to Polkadot-based asset transfers. Supports the project’s XCM-driven rights-transfer mechanisms, with a focus on event validation. Limited to liquidity focus, requiring adaptation for content rights.

6. **Yin, Z., et al. (2022).** Bool Network: An open, distributed, secure cross-chain notary platform. *IACR ePrint Archive*. <https://eprint.iacr.org/2022/1290.pdf>

Annotation: Proposes a cross-chain notary protocol with distributed key management. Relevant for secure rights transfers in the project, particularly for purchase verification. Limited by its focus on notarization over monetization models.

7. **Kuongho, C., et al. (2023).** A trusted reputation management scheme for cross-chain transactions. *Sensors*, 23(12), 6033. https://orbit.dtu.dk/files/330353284/sensors_23_06033.pdf

Annotation: Introduces a reputation scheme to ensure cross-chain security and prevent misbehavior. Supports the project's security evaluation, particularly for trustless PPV transactions. Limited to reputation systems, lacking direct monetization insights.

8. **Li, L., et al. (2024).** Why Blockchain-Based Digital Assets Are Owned on Decentralized Metaverse Platforms? *ScholarSpace*. <https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/da7d3ba9-379e-48a8-98e8-18b2bc293b1f/content>

Annotation: Explores decentralized ownership in metaverse contexts, which are relevant to the project's NFT-based purchase model. Supports Web3 principles but focuses on virtual assets, limiting its applicability to media-specific applications.

Content Rights Management Literature

9. **Wamugo, G. (2024).** Content monetization: non-fungible tokens as a new revenue stream for the media sector in Kenya. *Theses and Dissertations*. https://ecommons.aku.edu/cgi/viewcontent.cgi?article=3232&context=theses_dissertations

Annotation: Examines NFTs for media monetization, directly supporting the project's purchase verification model. Demonstrates revenue potential for creators but is limited to a Kenyan case study, requiring broader validation.

10. **Garba, A., et al. (2021).** A digital rights management system based on a scalable blockchain. *Peer-to-Peer Networking and Applications*, 14(5), 2665–2677. <https://research-api.cbs.dk/ws/portalfiles/portal/93314574/>

Annotation: Proposes a scalable blockchain DRM with watermarking for media transactions. Relevant for the project's scalability goals in subscription and PPV models. Limited by its single-chain focus, it lacks cross-chain perspectives.

11. **Mareckova, D. (2024).** Blockchain and Collective Rights Management of Copyright and Related Rights at the Global Level The Case of the Music Industry. *OAPEN Library*. https://library.oapen.org/bitstream/handle/20.500.12657/94575/oa_pdf-057-1731486746.pdf

Annotation: Explores blockchain for collective rights management, which is relevant to transparent revenue sharing in the project. Highlights creator empowerment but focuses on collective rights, limiting insights into individual monetization.

12. **Moncada, R., et al. (2021).** DLT-enabled Platforms: Distribution and Management of Media Content. https://mediaverse-project.eu/wp-content/uploads/2021/09/Research-Note-Blockchain-based-Platforms_integrated.pdf

Annotation: Discusses blockchain platforms for content distribution and royalties. Supports the project's PPV and subscription models with pay-per-use frameworks. Limited to European media, requiring global adaptation.

13. **Orero, P., et al. (2024).** The visible subtitle: Blockchain technology towards right management and minting. <https://open-research-europe.ec.europa.eu/articles/3-26/pdf>

Annotation: Focuses on blockchain for rights claiming and minting, supporting the project's purchase model with NFT-based ownership. Emphasizes scalability but lacks cross-chain specifics.

14. **Patil, S., & Dandge, P. (2022).** Blockchain-based multimedia content protection using encryption. *International Journal of Innovative Research and Technology*. https://ijirt.org/publishedpaper/IJIRT156400_PAPER.pdf

Annotation: Describes blockchain DRM for multimedia with encryption. Supports the project's security requirements for content access control. Limited by its focus on protection over monetization or cross-chain functionality.

15. **Buzu, I. (2021).** Blockchain, Smart Contracts and Copyright Management Disruption. *SSRN Electronic Journal*. https://papers.ssrn.com/sol3/Delivery.cfm/SSRN_ID3759260_code3591422.pdf

Annotation: Examines smart contracts for automated copyright management. Supports the project's ink! contract design for subscription automation. Limited to copyright, lacking PPV/purchase focus.

16. **Steem. (2025).** Steem, An incentivized, blockchain-based, public content platform. <https://steem.com/SteemWhitePaper.pdf>

Annotation: Describes a blockchain platform rewarding content creators. Relevant for the project's economic efficiency, demonstrating low-barrier monetization. Limited by its single-chain architecture.

17. **USPTO/USCO. (2024).** Non-Fungible Tokens and Intellectual Property — A Report to Congress. <https://www.copyright.gov/policy/nft-study/Joint-USPTO-USCO-Report-on-NFTs-and-Intellectual-Property.pdf>

Annotation: Examines NFTs for IP management, including licensing and ownership. Directly supports the project's purchase verification model, addressing legal implications. Limited to policy, not technical implementation.

18. Xie, R., & Zhang, W. (2022). Research on Copyright Protection of Digital Works Based on Multi-blockchain. *Recent Advances in Computer Science and Communications*, 15(8), 1126–1134. <https://doi.org/10.2174/2666255815666220204092947>

Annotation: Proposes a multi-blockchain framework for copyright protection, reducing block creation times. Provides benchmarks for the project's scalability evaluation. Limited by its focus on protection over monetization.

19. Xie, R., Tang, M. (2024). A digital resource copyright protection scheme based on blockchain cross-chain technology. *Heliyon*, 10(16), e34781. <https://doi.org/10.1016/j.heliyon.2024.e34781>

[g/10.1016/j.heliyon.2024.e34781](https://doi.org/10.1016/j.heliyon.2024.e34781)

Annotation: Details a cross-chain copyright scheme for registration and enforcement. Directly relevant for the project's rights transfer processes, improving single-chain limitations. Limited to protection-focused workflows.